

Name

- For the ODE $y' = Ay + \begin{pmatrix} e^{2t} \\ 3e^{2t} \end{pmatrix}$ with A below

In[*]:= MatrixForm[Eigensystem[$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$]]

Out[*]//MatrixForm=
 $\begin{pmatrix} 3 & 1 \\ \{1, 1\} & \{-1, 1\} \end{pmatrix}$

- Write down the gen sol.

$$y_c(t) = c_1 e^{3t} \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 e^t \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- Write down the form for the particular solution

$$y_p(t) = e^{2t} u$$

- Write down the Lin System for the coefficients

$$(2I - A)u = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

1. For the ODE $y' = Ay + \begin{pmatrix} e^{2t} \\ -e^{2t} \end{pmatrix}$ where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

- 1.1. Write down the form for the particular solution

$$y_p(t) = e^{2t} u$$

- 1.2. Write down the Lin System for the coefficients

$$(2I - A)u = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$y_p' = Ay_p + e^{2t} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$
$$2e^{2t}u = e^{2t}Au + e^{2t} \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$2u = Au + \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$(2I - A)u = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

$$y_p = e^{2t} u$$

$$2e^{2t}u = Ae^{2t}u + \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$(2I - A)u = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$y' = Ay + \sin(2t) \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

■ For the ODE $y' = Ay + \begin{pmatrix} \sin(2t) \\ -\sin(2t) \end{pmatrix}$ where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

■ Write down the form for the particular solution

$$y_p(t) = \sin(2t)u_1 + \cos(2t)u_2$$

■ Write down the Lin System for the coefficients

$\cos(2t)$ eq

$\sin(2t)$ eq

$$\begin{bmatrix} 2I & -A & | & 0 \\ A & -2I & | & -1 \end{bmatrix}$$

2. For the ODE $y' = Ay + \begin{pmatrix} t+1 \\ 2t \end{pmatrix}$ where $A = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$.

2.1. Write down the form for the particular solution

$$y_p(t) = u_0 + tu_1$$

2.2. Write down the Lin System for the coefficients

$$\begin{bmatrix} -A & I \\ 0 & -A \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 2 \end{bmatrix}$$

$$y_p' = Ay_p + \sin(2t) \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$2\cos(2t)u_1 + 2\sin(2t)u_2 = \cos(2t)Au_2 + \sin(2t)Au_1 + \sin(2t) \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$\cos(2t)$ eq.

$$2u_1 = Au_2$$

$$2u_1 - Au_2 = 0$$

$$2Iu_1 - Au_2 = 0$$

$\sin(2t)$ eq

$$2u_2 = Au_1 + \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$2Iu_2 - Au_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$Au_1 - 2Iu_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$y_p' = Ay + \begin{pmatrix} 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} u_1 \end{pmatrix} = A(u_0) + tu_1 + \begin{pmatrix} 1 \\ 0 \end{pmatrix} + t \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

const eq

$$u_1 = Au_0 + \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \boxed{-Au_0 + Iu_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}}$$

"t" eq

$$\begin{pmatrix} 0 \end{pmatrix} = Au_1 + \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$0u_0 - Au_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

If you prefer the Aug Mat version

$$\begin{bmatrix} -A & I & | & 1 \\ 0 & -A & | & 2 \end{bmatrix}$$